

Assignment: Fundamentals of Linear Algebra for Artificial Intelligence

Instructions

- Answer all the questions.
 - Show all intermediate calculations wherever applicable.
 - Use a calculator if required.
 - Round numerical answers to **two decimal places** where necessary.
 - Clearly write the final answer for each question.
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Part A – Scalars, Vectors and Matrices

Question 1

Identify whether each of the following is a **Scalar**, **Vector**, **Matrix**, or **Tensor**.

a)

$$5$$

b)

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

d)

A colour image of size

$$256 \times 256 \times 3$$

Answer- A

Question 2

Determine the dimensions of the following matrices.

a)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

b)

$$\begin{bmatrix} 5 \\ 8 \\ 9 \\ 2 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Answer- A-(2*3), B-(4*1), C-(3*2)

Question 3

Given

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

Find

- $A + B = \begin{bmatrix} 6 & 6 \\ 9 & 15 \end{bmatrix}$
 - $A - B = \begin{bmatrix} -2 & 6 \\ -3 & -1 \end{bmatrix}$
 - $2A = \begin{bmatrix} 4 & 10 \\ 6 & 14 \end{bmatrix}$
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Question 4

Find the transpose of the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}$$

8 10 12

$$\text{Ans-} \begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$$

Question 5

Find the transpose of

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix}$$
$$\text{Ans-} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \end{bmatrix}$$

Part B – Matrix Multiplication

Question 6

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$\begin{aligned} A * B &= \begin{bmatrix} 1*5+2*7 & 1*6+2*8 \\ 3*5+4*7 & 3*7+4*8 \end{bmatrix} \\ &= \begin{bmatrix} 20 & 22 \\ 43 & 53 \end{bmatrix} \end{aligned}$$

Question 7

Determine whether the following matrix multiplications are possible. If they are possible, write the dimensions of the resulting matrix.

a)

$$(2 \times 3)(3 \times 4)$$

Ans- Yes

b)

$$(3 \times 2)(4 \times 3)$$

Ans- No

c)

$$(4 \times 2)(2 \times 5)$$

Ans- Yes
d)

$$(5 \times 4)(4 \times 2)$$

Ans- Yes

Question 8

Given

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Find

$$AB$$

Ans-

$$\begin{bmatrix} 17 & 20 \\ 41 & 48 \end{bmatrix}$$

Question 9

Verify whether matrix multiplication is commutative.

Compute

$$AB$$

and

$$BA$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Based on your calculations, what do you conclude?

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \quad BA = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$$

Since – $AB \neq BA$

Part C – Vector Operations

Question 10

Find the magnitude of the vector

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Ans- Magnitude -6

Question 11

Find the magnitude of

$$y = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

Answer- ~8.06

Question 12

Compute the dot product of

$$x = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$$

Ans- $x \cdot y = 21$

Question 13

Determine whether the following vectors are orthogonal.

$$u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$
$$v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Ans-

$$u \cdot v = (2 * 1) + (1 * -2)$$

$$u \cdot v = 2 + (-2)$$

$$u \cdot v = 0$$

Since the dot product is 0, the vectors are orthogonal.

Question 14

Normalize the vector

$$x = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

Ans-

$$||x|| = \sqrt{6^2 + 8^2}$$

$$||x|| = \sqrt{36 + 64}$$

$$||x|| = \sqrt{100}$$

$$||x|| = 10$$

$$\text{Normalized vector} = \begin{bmatrix} 6 / 10 \\ 8 / 10 \end{bmatrix}$$

$$\text{Normalized vector} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

Question 15

Find the Euclidean distance between

$$A = (2, 3)$$

and

$$B = (8, 11)$$

$$\text{Ans- } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = 10$$

Part D – Vector Spaces and Span

Question 16

Determine whether the following vectors are linearly independent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Explain your answer.

$v_2 = 2 * v_1$, the second vector is a direct scalar multiple of the first vector. Therefore, the vectors are **linearly dependent** (they are NOT linearly independent)

Question 17

Determine whether

$$v = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

belongs to the span of

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$v_2$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

vector v belongs to the span of v_1 and v_2 if it can be written as a linear combination of them. This means we need to find scalars c_1 and c_2 such that:

$$v = c_1 * v_1 + c_2 * v_2$$

1. Set up the system of equations:

$$\begin{bmatrix} 5 \\ 7 \end{bmatrix} = c_1 * \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 * \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

This gives us two individual linear equations:

$$\text{Equation 1 (Top components): } c_1 + 2c_2 = 5$$

$$\text{Equation 2 (Bottom components): } c_1 + c_2 = 7$$

2. Solve the system of equations:

Subtract Equation 2 from Equation 1 to eliminate c_1 :

$$(c_1 + 2c_2) - (c_1 + c_2) = 5 - 7$$

$$c_2 = -2$$

Now substitute $c_2 = -2$ back into Equation 2 to find c_1 :

$$c_1 + (-2) = 7$$

$$c_1 = 9$$

3. Verify the solution:

Check if $c_1 = 9$ and $c_2 = -2$ works for Equation 1:

$$9 + 2(-2) = 9 - 4 = 5 \text{ (True)}$$

4. Conclusion:

Since a valid solution exists ($v = 9v_1 - 2v_2$), the vector v **does belong** to the span of v_1 and v_2 .

Question 18

Find a basis for

$$\mathbb{R}^2$$

Ans-

$$V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Question 19

Find the rank of the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Ans- Rank- 1

Question 20

Find the rank of

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Rank- 2

Part E – Orthogonality

Question 21

State whether the following pairs of vectors are orthogonal.

a)

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Ans- Yes (dot product is zero)

b)

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

Ans- Yes (dot product is zero)

c)

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Ans- No(Dot product is 20 0)

Question 22

Find the cosine similarity between

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Interpret the result.

Ans-

$$x \cdot y = (1 * 3) + (2 * 4)$$

$$||x|| = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5} \approx 2.236$$

$$||y|| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\text{Cosine Similarity} = 11 / (\sqrt{5} * 5)$$

$$\text{Cosine Similarity} \approx 0.984$$

Question 23

Explain, in your own words, the difference between:

- Orthogonal vectors - A set of vectors that are entirely perpendicular to each other
- Orthonormal vectors- they point at 90-degree angles to one another and are all exactly 1 unit long.
- A basis- A minimal set of building-block vectors used to construct a specific mathematical space.
- A vector space- It is a complete collection of vectors that follows specific algebraic rules (like being able to add vectors or scale them without leaving the space).

